

RAKESH REDDY NALLAGONDLA

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience in building and optimizing **ETL / ELT** workflows, managing big data, and delivering scalable cloud-based data solutions across **AWS** and **Azure** platforms. Skilled in designing efficient data pipelines, ensuring data governance, and enabling real-time data processing using tools like **Apache Kafka**, **Amazon Kinesis**, and **Azure Event Hubs**. Proficient in **data modeling with dbt**, and experienced with **Snowflake**, **Amazon Redshift**, and **Azure Synapse Analytics** for robust data warehousing and analytics. Hands-on with processing diverse datasets including healthcare data (**HL7v2**, **FHIR**) and **IoT** telematics data, with strong focus on data quality, validation, and regulatory compliance (**GDPR**, **HIPAA**, **SOX**). Adept at translating business needs into scalable solutions using **Azure Data Factory (ADF)**, **AWS Glue**, and orchestration tools like **Apache Airflow**. Strong problem-solving skills with proven collaboration across cross-functional teams to align data strategy with business goals. Experienced in **CI/CD** automation using **GitHub Actions**, **Azure DevOps**, and tools like **Docker**, **Jenkins**, and **Terraform**, ensuring smooth deployment and system reliability.

TECHNICAL SKILLS

- **Programming & Scripting:** Python (NumPy, Pandas, Scikit-learn, Boto3), SQL, Java, R, Linux/UNIX, Scala
- **Cloud Platforms:** AWS (S3, EC2, Redshift, Glue, Lambda, Kinesis, RDS, Athena, Step Functions, SQS/SNS, EKS, VPC), Azure (Data Factory, Databricks, Synapse, Cosmos DB, Event Hubs, Azure SQL, Data Lake, Delta Lake), GCP (BigQuery, Dataflow, Pub/Sub, Google Analytics)
- **Data Warehousing & Databases:** Snowflake, Amazon Redshift, DynamoDB, Azure Synapse, PostgreSQL, Cassandra, Cosmos DB, MySQL, Oracle PL/SQL, SSIS, SSMS, NoSQL
- **Data Engineering Tools:** Apache Spark (PySpark), Kafka, Apache Airflow, Hive, Apache Flink, Informatica PowerCenter, SnowSQL, BigData, Hadoop, Linux, PowerShell, CRON jobs
- **Data Formats:** JSON, Parquet, Avro, ORC, XML, CSV, YAML
- **Visualization Tools:** Power BI, Tableau, Python (Matplotlib, Seaborn)
- **Enterprise Tools & Platforms:** Salesforce, SAP
- **CI/CD & DevOps:** Jenkins, GitHub, GitLab, Git, Azure DevOps, Docker, Kubernetes, Terraform
- **IDEs & Development Tools:** PyCharm, Anaconda, Visual Studio Code, Jupyter Notebooks
- **Monitoring & Logging:** AWS CloudWatch, Azure Monitor, Log Analytics
- **Project & DevOps Tools:** JIRA, Bitbucket, Confluence, Agile, Scrum, Kanban

PROFESSIONAL EXPERIENCE

Client: Fidelity Investments

Data Engineer | Westlake, TX | Feb 2024 – Present

- Migrated legacy ETL workflows from Informatica PowerCenter to AWS Glue (PySpark), improving pipeline efficiency, scalability, and **reducing processing time by 30%**.
- Automated data ingestion and transformation workflows using **AWS Lambda**, **Glue**, and **Step Functions**, efficiently processing large volumes of structured and semi-structured data from multiple sources.
- Integrated **Snowflake** as the centralized data warehouse, enabling high-performance querying and reporting; utilized **dbt** for developing modular, reusable, and version-controlled **data models**.
- Designed and developed **REST APIs** to support real-time data ingestion, integrating with **Amazon Kinesis** for streaming data into the system with minimal latency.
- Performed comprehensive data **transformation and validation** in **AWS Glue (PySpark)**, implementing schema checks and data cleansing routines to ensure accurate and reliable datasets for analytics.
- Managed and optimized **Amazon Redshift** for specific reporting workloads, supporting ad-hoc analysis and historical data storage.
- Leveraged **Athena** for ad-hoc querying of raw data in S3 and integrated **SQS/SNS** for event-driven data processing workflows.
- Orchestrated and scheduled end-to-end data workflows using **Apache Airflow**, improving automation, monitoring, and reliability of data pipelines.
- Monitored pipeline health using **AWS CloudWatch**, configuring metrics, logs, and alarms for real-time system performance tracking and proactive issue resolution.
- Automated CI/CD workflows using **GitHub Actions** and **AWS CodePipeline**, integrating unit testing and deployment approvals for production readiness.
- Implemented robust **data security** by configuring **IAM roles**, **KMS encryption**, and access controls, ensuring compliance with **GDPR** and **SOX** regulatory standards.
- Actively collaborated with **cross-functional teams** including data analysts, architects, and business stakeholders to optimize end-to-end data workflows.

Client: Seattle Children's

Data Engineer | Seattle, WA | Jun 2021 – Dec 2022

- Designed and implemented **Azure Data Factory (ADF)** pipelines to ingest and orchestrate the flow of **HL7v2 healthcare data** from multiple sources into **Azure Data Lake Storage**, ensuring secure and efficient data ingestion at scale.
- Engineered robust **ETL workflows in Azure Databricks (PySpark)** to transform **HL7v2 messages into FHIR-compliant resources**, supporting downstream healthcare analytics and interoperability requirements.
- Performed **schema validation and quality checks** on transformed data to ensure **FHIR compliance**, **data integrity**, and readiness for **analytics in Azure Synapse Analytics**.

- Managed both structured and semi-structured data storage in **Cosmos DB** and **Azure SQL Database**, adhering to **HIPAA-compliant** standards, with focus on **low-latency access** and **data security**.
- Developed **interactive dashboards in Power BI**, integrated with **Azure Synapse Analytics**, delivering actionable insights and reports to clinical and administrative teams in real time.
- Streamlined FHIR pipeline delivery with **Azure DevOps CI/CD**, leveraging pipeline templates and infrastructure-as-code (**Terraform**) for version control and reproducibility.
- Configured custom alerts in Azure Monitor and Log Analytics, **reducing incident response time and ensuring 99.9% pipeline uptime**.
- Worked collaboratively with **clinical informatics teams** and data stakeholders to **map, validate, and refine healthcare data models**, resulting in improved **data accuracy**, reporting quality, and alignment with **FHIR standards**.
- Ensured **data governance** by implementing role-based access control (**RBAC**) via **Azure Active Directory (AAD)**, along with encryption and audit logging for secure data handling and **regulatory compliance**.

Client: Progressive Insurance

Junior Data Engineer | Mayfield Village, OH | Jan 2020 – May 2021

- Developed and maintained **Azure Data Factory (ADF) pipelines** to ingest and process high-volume **telematics (IoT device) data** collected from vehicles for auto insurance analytics.
- Engineered **ETL workflows in Databricks (PySpark)** to process high-volume **IoT data**, applying validation rules for accurate analysis and reporting.
- Optimized Azure Synapse Analytics for efficient querying, **cutting report generation time by 25%** and enhancing stakeholder decision-making.
- Managed real-time and historical data storage using **Cassandra** and **Cosmos DB**, ensuring **low-latency access** and **high availability** for streaming and batch data use cases.
- Implemented **data validation** rules and quality checks during transformation to detect anomalies in telematics data, improving reliability of insights used by analytics teams.
- Set up end-to-end **monitoring and alerting** for pipelines using **Azure Monitor** and **Log Analytics**, enabling proactive performance tracking and quick issue resolution.
- Automated code integration and deployments with **Azure DevOps CI/CD pipelines**, reducing manual intervention, deployment time, and increasing release consistency.
- Applied **role-based access control (RBAC)** via **Azure Active Directory (AAD)** to safeguard sensitive insurance data, ensuring compliance with security standards.
- Collaborated closely with **data analysts and risk management teams** to deliver clean, validated datasets for real-time **risk analysis, usage-based insurance (UBI)** models, and business reporting.
- Ensured adherence to **data governance policies**, contributing to the development of standardized data processing frameworks and documentation for long-term maintainability.

EDUCATION

Master of Science | Applied Statistics and Decision Analytics

Western Illinois University | Jan 2023 – Dec 2024 | GPA: 3.8/4.0

Key Courses: Introduction to Probability and Statistics, Data Visualization with Tableau, Big Data for Business Decision Making, Applied Data Mining for Business Decision Making, Categorical Data Analysis Using Logistic Regression, Predictive Analytics.

ACADEMIC PROJECTS

Predictive Analytics for Business Decision Making

- Developed predictive models using **Python (Scikit-learn, Statsmodels)** to forecast sales trends and customer behavior.
- Applied regression analysis and machine learning to support optimized business decisions.
- Built interactive dashboards using **Tableau** for data visualization.

Categorical Data Analysis & Logistic Regression

- Conducted logistic regression to predict customer retention patterns.
- Performed hypothesis testing using **Python (Statsmodels, Scipy, Pandas)**.
- Visualized findings using **Matplotlib** and **Seaborn** for insights.